B51TF Availability – Design Exercise (20%)

Note that it is against university policy to **collude** with other students in this assessment activity. The repercussions for doing this can be high. All material used / presented from other sources should be suitably referenced.

Assessment Details

It is proposed to design a small-scale system for using high-pressure air as an energy storage system. See the diagram below. The system consists of a compressor connected to a pressure vessel. Between the compressor and the pressure vessel is a heat exchanger that is used to supply hot water. The pressure vessel is also connected to a turbine. The valve between the pressure vessel and the turbine is a flow control valve. In a flow control valve, provided that the pressure is above a specified minimum value the mass flow is fixed. Once the pressure in the vessel reaches the minimum value, the valve closes.

The logic of the design is that a renewable energy system, such as a system of solar photovoltaic cells (PV) or a wind turbine, powers the compressor throughout the day. During this activity, some of the power is used to supply hot water, and the rest to fill the pressure vessel. When power is required the high-pressure air is used to drive a turbine delivering electricity as required.

Note: The system is to be considered for summer or winter conditions only.

Simplifications

- The valves are not to be analysed as part of the availability analysis.
- Calculate the temperature and pressure of the vessel after filling based on a closed system analysis for the air entering the cylinder and the air already in the cylinder.
- Assume that the temperature of the vessel remains constant during fluid outflow to the turbine.

Conditions Imposed

The compressor inlet temperature, $T_{in,\ comp}$, is to be determined using the last two digits of your student ID. See the equation below

Equation:

S = last two digits of your students ID.

- If student ID ends up in 00, then *S* = 0;
- if student ID ends up in 34 then *S* = 34.

$$T_{in,\ comp} = 5 + 20 * (S / 100) \ \ [\text{in } ^{o}\text{C}]$$

Summer or Winter Conditions

Property	**Value**
Compressor	
$P_{in,\ comp}$	1 bar
$T_{in,\ comp}$	See the equation above
$\dot{W}_{in}$ (supplied by PV cells or wind power)	Check the Allocation List on Canvas
t_{comp}	Check the Allocation List on Canvas
Heat Exchanger	
Water Stream	
$T_{in,\ w}$	Check the Allocation List on Canvas
$T_{out,\ w}$	80ºC
c_w	4.2 kJ / kg K
Turbine	
Isentropic Efficiency	0.85
$P_{out,\ turb}$	1.1 bar

Conditions to be prescribed

Property	Range of possible values
Compressor	
$P_{out,\ comp}$	Student to prescribe
Heat Exchanger	
$T_{out,\ air}$	Student to prescribe
Pressure Vessel	
V_{vessel}	Student to prescribe between 50m^3-300m^3
$P_{vessel,\ min}$	Student to prescribe between 2 bar – 15 bar
Turbine	
t_{turb}	Student to prescribe between 1 hour / day – 4 hours / day

Note all thermal devices MUST satisfy the 1st and 2nd law of thermodynamics and conservation of mass.

"Student to prescribe" = the value is specified by the student. All other values can be derived from a thermodynamic analysis and appropriate assumptions as required.

Starting point for your analysis

Probably the best place to start is to state the vessel volume, $V_{vesssel}$, the pressure at the outlet of the compressor and the minimum pressure of the vessel.

The mass flow rate of the compressor and the turbine are determined as,

$$\dot{m}_{comp} = \frac{M_{max} - M_{min}}{t_{comp}}$$

$$\dot{m}_{turb} = \frac{M_{max} - M_{min}}{t_{turb}}$$

Where M_{max} and M_{min} are the maximum mass in the vessel and the minimum mass in the vessel based on the minimum pressure, (setting for the flow control valve).

The rest is up to you.

Activities

- For a set of conditions, complete an availability analysis for each thermal unit (compressor, heat exchanger, and turbine but NOT the pressure vessel).

- Design a system that will deliver an acceptable balance of water heating and power delivery. State what this system might be used for – e.g., 1 house, a row of houses, a town, a commercial operation, a country! Explain and demonstrate how your system satisfies the brief and is optimal. You may consider the vessel size, the compressor outlet pressure, and the vessel minimum pressure to optimise your design.

- You are strongly encouraged to use a spreadsheet for both parts to explore parameter space.
- Please submit a report and spreadsheet (if produced) to Canvas, combining both parts in one report.

Revision

Some of your analysis will rely on 2nd and 3rd year thermodynamics such as the use of isentropic efficiency, for example. You are strongly advised to refresh your memory of this aspect of thermodynamics using the screencasts revising this material and consult your own notes.

Nomenclature

P_0, T_0	- Ambient pressure and temperature
$\dot{m}_{comp}$	- Mass flow rate of air into the compressor
$P_{in,\ comp}$	– Inlet pressure to compressor
$T_{in,\ comp}$	– Inlet temperature to compressor
$P_{out,\ comp}$	– Outlet pressure to compressor
$T_{out,\ comp}$	– Outlet pressure to compressor
$\dot{W}_{in}$	– Work powering compressor
t_{comp}	-Time interval in a day the compressor is running
$T_{in,\ w}$	- Water stream inlet temperature to heat exchanger
$T_{out,\ w}$	- Water stream outlet temperature to heat exchanger
$T_{out,\ air}$	- Air stream outlet temperature to heat exchanger
V_{vessel}	- Total volume of pressure vessels
$P_{vessel,\ max}$	- Maximum pressure of vessel
$P_{vessel,\ min}$	- Minimum pressure of vessel set by flow control valve
T_{vessel}	- Temperature of pressure vessels
M_{min}, M_{max}	- Minimum and maximum mass of air in the vessel
$\dot{m}_{turb}$	- Mass flow rate of air in the compressor
$\dot{W}_{out,turb}$	- Work out of turbine
t_{turb}	-Time interval in a day the turbine is running

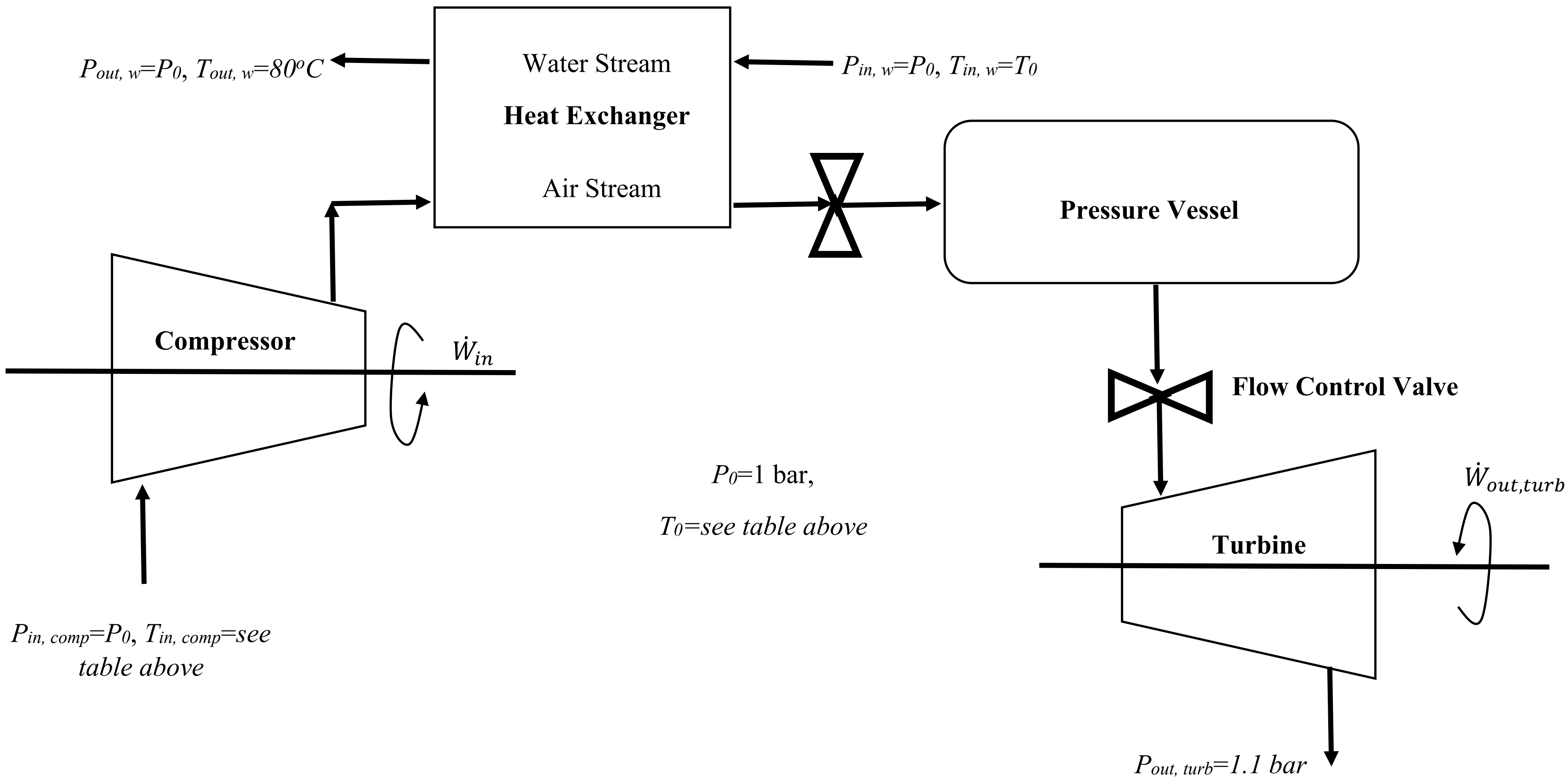

Figure 1 - Thermal / Working Devices - Renewable Energy Storage and delivery